Lesson 8 Teacher's Key: Under Pressure

001 Game Creator (Windows only) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Adding a second consequence to an existing Trigger ("Left/Entered Camera," built in Lesson 7 for escape cleanup) - reinforcing exactly why last week's fix (splitting into two specific Triggers) mattered: this week's addition was trivial to make correctly BECAUSE the escape Trigger was already isolated from the hit Trigger.

Level 1 - Full Worked Solution

- lives is a new Variable, parallel to enemies_remaining - same pattern, different variable, different Trigger responsible for changing it.
- The lives decrement and Message Box are added directly inside the EXISTING 'Left/Entered Camera' Trigger from Lesson 7, BEFORE the existing Destroy event - new Events added to an already-correct Trigger, not a new Trigger.
- 'Collided with actor' remains completely untouched this week - it's still the only place enemies_remaining changes.

Highlighted lines are new this week; everything else is code you already have.

```
enemy Actor Trigger: Left/Entered Camera (Left case) -> (revised)
  Variable – Modify Value(lives, Subtract, 1)
  If(lives <= 0)
    Message Box("Game Over")
  Actor – Destroy/Revive Actor(this, Destroy)
```

Why This Structure

- This is the most direct possible payoff of Lesson 7's fix: because 'Left/Entered Camera' was already the single, unambiguous home for 'this enemy left the screen unshot,' adding a new consequence to it required touching nothing else.
- Contrast this with what would have happened if Lesson 7's fix hadn't been made (i.e. if 'Defeated / Destroyed' were still doing double duty): this week's lives logic would have had to be untangled from the old broken win-counter at the same time as being added.

Level 2 / Level 3 Talking Points

- Level 3 sound, worked (001 Game Creator): import both files. In the enemy Template's "Left/Entered Camera" Trigger (Left case): after the lives change, add Play Sound (sfx_lifelost.wav); inside the lives = 0 check, add Play Sound (sfx_gameover.wav) before the Message Box.

- This is a great lesson to explicitly connect back to last week: ask students to recall why the 001 Game Creator fix mattered, and now show them it's the reason this week was easy.
- Level 3 structure: the enemy Actor Template now has exactly two Triggers doing the entire win/lose logic of the game so far - worth pointing out how compact that still is, even as features accumulate.

Level 2 - complete code:

```
// Level 2 is verification only - the Level 1 code is already correct, so
no lines change.
// Checks: shooting an enemy never changes lives (only the kill path
changes enemies_remaining);
// an escape never changes enemies_remaining (only the escape path changes
lives);
// "Game Over" appears once, when lives first reaches 0.
```

Level 3 - complete code (all three goals combined):

Highlighted lines are the Level 3 additions on top of the Level 1 code.

```
enemy Actor Trigger: Left/Entered Camera (Left case) -> (Level 3)
  Variable – Modify Value(lives, Subtract, 1)
  Play Sound(sfx_lifelost.wav)
  If(lives <= 0)
    Play Sound(sfx_gameover.wav)
    Message Box("Game Over")
  Actor – Destroy/Revive Actor(this, Destroy)

Level 3 visual: add a Text/HUD element bound to the lives variable and
refresh it in this same trigger.
```

Common Student Mistakes

- Adding the lives decrement to 'Collided with actor' instead of 'Left/Entered Camera' (makes shooting cost a life, which is backwards).
- Forgetting lives starts at 3 (check the Variable's default value in the Variables list, not just referenced in an Event).
- "Game Over" and "You Win!" Message Boxes both appearing if an enemy escapes at the same moment the last enemy is shot - an edge case, not a bug, worth a quick acknowledgment if a student hits it.